GLOBAL
; up · **.** down · **,** **/** left/right · **ENTER** select · **`/DEL** back · Home: 2-col grid, letter jumps, **,/** hop columns · **{ }** page · **b** screenshot · **h** help · **Fn+`/Fn+DEL** STOP all radio/rotator control (Help links: **g** glossary+math · **m** user guide · **s** sat history · **t** tech help · **l** learn theory · **f** band plan)

HOME
ENTER opens item; menu scrolls: Satellites, Next Passes, Passes, Track, World Map, Overhead now, Sun/Moon, Space Wx, Activations, AMSAT status, Weather, Grid dist/bearing, QRZ Lookup, Location, Update, Settings, Log, Messages, About, Charge/Sleep

SATELLITES
f favorite · **v** favs-only · **/** search ALL of CelesTrak → add as auto-updating fav (10 s/2 h courtesy limits) · **n** new GP sat · **x** del added sat · **e** EQX table · **k** OSCARLOCATOR · **3** 3D globe · **2** sat-to-sat · **o** orbital · **y** sim · **t** transponders · **d** 10-day · **i** illum · **s** AMSAT status · **L** share GP over LoRa · **ENTER** passes · right edge: dot = AMSAT heard, square = telemetry, ring = not heard

SAT-TO-SAT (Sats → 2)
Windows when selected sat + a 2nd fav are BOTH above your horizon (start, dur, peak el each). **n** next sat · **r** recompute · ` back

LIVE GPS POS (Location → v)
DMS lat/lon (max precision) + decimal, alt, grid, speed, course. Rovers/portable. ` back

3D GLOBE (Sats → 3)
Wireframe Earth, auto-follows selected sat. Graticule, coastline, day/night terminator, QTH, favs, selected sat + footprint + ground-track trail. **g** DX 2nd location (grid) + footprint · **G** clear · **arrows** turn · **ENTER** re-follow · ` back

EQX TABLE (Sats → e)
Equator-crossing UTC + longitude (W-positive) for OSCARLOCATOR, next 3 days. **;/.** scroll · **d** asc/desc node · **r** recompute · ` back

OSCARLOCATOR (Sats → k)
Live azimuthal-equidistant plot: sub-point, footprint, QTH range ring + full ground track (AOS/LOS). **m** toggle polar (default, auto N/S, flips at equator) / QTH-centered · ` back

DX DOPPLER TABLE (Mutual → ENTER → d)
RX/TX dial freqs for BOTH stations every 30s across a mutual window. Two lines/step: me (green) + DX (cyan). From the Mutual list, **ENTER** opens a polar detail (me+DX arcs, AOS/LOS/el) then **d** the table (or **d** straight from the list). **t** cycle transponder · **m** mode: true rule / fixed DL / fixed UL · **a** anchor dial (me/DX RX/TX) · **,/** in fixed mode step anchored dial to round 1 kHz (else passband 1 kHz) · header shows pb +/- from center · **;/.** scroll · ` back

NEXT PASSES (favs)
ENTER track · **m** world map · **t** sky-at-a-glance timeline (bars by elev) · **p** rove planner · **w** workable horizon · **s** target search · **r** refresh · **z** deep-sleep until AOS

WORKABLE HORIZON (Next Passes → w)
10-day union of everything workable across all favorites. **s** states list · **d** DXCC list · **g** re-run incl. grids (off by default) · **w** save to /CardSat/workable/ · ` back

TARGET SEARCH (Next Passes → s)
Pick one state / DXCC / grid; lists every pass over 10 days where it's workable, time-ordered across all favs (sat / date / window / maxEl). **ENTER** polar of that pass · **w** save to /CardSat/search/ · ` back

ROVE PLANNER (Next Passes → p)
Survey all favorites from a planned grid+time. Form: grid / date / time / +/- hrs / GO. Rows by AOS: sat / AOS / maxEl / states / DXCC. **ENTER** detail (arc + **s/d/g** state/DXCC/grid lists) · **w** save txt · **l** saved plans (**ENTER** view, **;/.** scroll, **d** del) · **g** form

PASSES (sel)
***** = optically visible. **;/.** select · **d** detail · **t/ENTER** track · **n** add TX · **r** recompute · **x** mutual-DX · **v** 10-day chart · **V** visible-pass list · **i** illum · **g** grids · **w** US states · **e** DXCC · **p** print

TRACK (sel)
` exits to previous screen & KEEPS radio/rotator tracking (green RAD/ROT/R+R in header); **r/o** to stop. **m** TUNE/CAL · **d** tune mode (FULL/DL/UL/FULLu/hold; FULLu=OTR on uplink knob) · **t** next TX · **n** jump to beacon · **c** CTCSS · **N** sat note · **k** CW both legs (linear) · **r** radio · **o** rotator · **p** polar · **a** point-here arrow · **z** big readout · **y** tilt on/off (ADV) · **f** Manual · **l** log QSO · **v** voice memo (SD) · **g** grids · **w** states · **e** DXCC now · **i**×2 report Heard (AMSAT) · after LOS **q** (60s) deep-sleep to next pass · **ENTER** save cal

BIG READOUT (z from Track)
Big RX/TX + az/el + tune mode (follows Track). Radio+rotator keep tracking · **,/** tune · **s/x** step/ctr · **m/d** mode · **t** TX · **n** beacon · **k** CW (linear) · **r** radio · **o** rot · **y** tilt · **l** log · **z/`** back

MANUAL (no radio)
Fix one leg, read Doppler freq to tune the other by hand. **u** toggle HOLD/TUNE leg · **,/** passband (linear) · **m** CAL · **t** next TX · **z** big view · **l** log · **p** polar · **g** grids · **w** states · **e** DXCC · **`/f** Track

MANUAL BIG (z from Manual)
HOLD/TUNE legs in big digits · **u** swap leg · **,/** tune · **s/x** · **m** CAL · **t** TX · **z/`** back

TRACK · TUNE
,/ tune -/+ · **s** step 100/1k/5k · **x** recenter · tilt to tune if enabled (ADV)

TRACK · CAL
,/ downlink · **;/.** uplink · **s** step 10/100/1k · **x** zero

WORKABLE GRIDS / STATES / DXCC
Footprint coverage (per-pass union or live now), count on a cyan line: **g** 4-char grids · **w** US states+DC · **e** DXCC (all 340, hybrid polygons+points). On grids, **f** prefix filter (EM/EM2/EM21, upper-cased), **c** clear. **;/.** & **{/}** scroll · ` back

POLAR / PASS DETAIL
Pass detail: **p** polar of this pass. Polar: **l** log QSO · **v** voice memo (SD) · **p/ENTER/`** back

SUN / MOON
Graphic sky-dome (Sun/Moon glyphs by az/el) · **g** graphic/list · **;/.** pick Sun/Moon · **o** rotor track on/off · **s** sky sources · **t** transits · **e** EME · **x** stop · ` back

SKY SOURCES (Sun/Moon → s)
Planets (cyan dots) + strong radio sources (orange +): Cas A, Cyg A, galactic center, Crab, Virgo A, on a sky dome. Antenna-pointing / RF reference. **;/.** select · ` back

EME / MOONBOUNCE (Sun/Moon → e)
Self-echo Doppler per band (50/144/432/1296/10368, topocentric) · range + rate · path degradation vs perigee · galactic sky-noise flag · SUN flag <10° · **m** mutual window · **p** 30-day plan · **o** rotor track Moon · ` back

GRID DIST/BEARING (menu)
Enter Maidenhead grid → great-circle distance + beam heading (short/long path, km/mi). **g** grid · **q** QRZ→grid lookup (seeds calc) · **o** point rotor at bearing · ` back

SPACE WX (menu)
Solar 10.7cm flux + planetary Kp + A index + aurora likelihood (from Kp), labeled & color-coded, with HF/sat operating outlook & data age · **p** HF/6m propagation · **m** MUF to world regions · **r** refresh (WiFi) · ` back

HF/6m PROPAGATION (Space Wx → p)
Turns solar flux + Kp into band guidance: HF conditions (10/15/20m open/marg/shut), geomagnetic effect, auroral-VHF likelihood (6m/2m, beam N), D-layer absorption. Rule-of-thumb · **r** refresh · ` back

MUF TO REGIONS (Space Wx → m)
MINIMUF-3.5 path MUF from your QTH to 24 world DX regions: bearing, distance, MUF, best band, color-coded. F-region model, best 800-8000km · **;/.** scroll · **k** world map (colored dots) · **x** print · ` back

WEATHER (menu)
Current conditions + multi-day forecast for your site (Open-Meteo). Refreshes on entry (WiFi) & with Update. Units in Settings · **r** refresh · cached offline · ` back

QRZ LOOKUP (menu)
Callsign lookup via QRZ.com XML (needs QRZ XML subscription + user/pass in Settings → Network). **ENTER** type call → name/addr/grid/class. WiFi req'd · ` back

BAND PLAN (Help → f)
Worldwide amateur band reference LF→light: HF with ITU R1/R2/R3 splits, VHF/UHF/microwave EME+calling freqs, satellite subbands, IARU designators (H/A/V/U/L/S/C/X/K), sat modes incl QO-100. **;/.** scroll · ` back

ACTIVATIONS (menu)
Upcoming sat activations on hams.at (roves, grid/special ops). List: date, call, sat, grid (* = your own entry). **;/.** move · **ENTER** detail · **n** add your own sked (offline OK), **e** edit a * entry · **r** refresh · ` back. Detail: UTC/mode/freq + **footprint note** (checks co-visibility with the activator +/-30 min of listed time), **;/.** scroll the full comment, **a** SKED reminder (T-60/30/10 beeps+flash), **w** mutual-window screen if a footprint exists. Mutual window: small polar plot (me + DX arcs), Date/AOS/LOS/dur + peak el each; **d** DX Doppler pre-set to the transponder & fixed DL/UL parsed from the freq/comment (default table if none). Cached to card — last list shows offline; WiFi to refresh

OVERHEAD NOW (menu)
Snapshot of every catalog sat above the horizon right now, sorted by elevation, with az + rise compass dir (high=green, low=yellow) + count up/scanned. **r** rescan (this instant) · **;/.** scroll · ` back

TRANSPONDER DB (Sats → t)
Sat's transponder/beacon entries, two lines each: **D**=downlink+mode line, **U**=uplink+tone/inv line. Ordered two-way > amateur > active; inactive dimmed & **(off)**. **;/.** select (* = manual) · **x** del manual (2x) · ` back

ORBITAL ANALYSIS
,/ 11 pages: Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position / Phys (velocity + launch date/age) / Explore (what-if apo/peri/incl sandbox, **x** reseed) · **r** recompute · **z** orbital-zone transits · Doppler **f** sets beacon freq

ORBITAL ZONES (Orbital → z)
When the sat transits distinctive regions, w/ enter/exit times + duration + live in/out + L-shell. **z** cycle zone: **SAA** / **Eclipse** / **Polar** (>60°) / **Inner** & **Outer Van Allen belts** (L-shell, higher orbits) · **;/.** scroll · **x** print · ` back

SIMULATION
,/ step time · **;/.** step size · **m** world-map view (sub-point + footprint) · **x** reset to now · ` back

LOCATION
e/o/a lat/lon/alt · **g** grid · **p** GPS on/off · **s** GPS source · **c** set clock · **v** live position (DMS) · **ENTER** GPS sky plot

GPS SKY PLOT
Live GNSS by az/el, colored by signal (green=strong, gray=weak) · ` back

WORLD MAP
All footprints + night-side shading · **f** highlight favorite · **c** recenter on QTH/0° · yellow=sunlit cyan=eclipse · ` back

SCHEDULES
10-day: **;/.** scroll +/-1 day (fills full days), **r** recompute · Illum: **,/** scroll +/-60 days · Mutual: **;/.** scroll, **ENTER** polar detail, **d** Doppler

LOG
Menu (scrolls): **ENTER** new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards / fill grids (QRZ) · **Awards**: QSO/grid/state/DXCC totals (states+DXCC derived from grid), **g**/**s**/**d** scrollable worked lists, **ENTER** per-sat · List: **;/.** scroll, **ENTER** edit · Entry: **;/.** field, **ENTER** edit, **s** save, **x** x2 delete · editing a QSO re-arms its upload; extra **LoTW**/**Cloudlog** rows (ENTER toggles) override that

SIGN & UPLOAD TO LoTW (Log → Sign & upload)
Uploads sat QSOs direct to ARRL LoTW over WiFi. Needs SD + your LoTW key (lotw_key.pem/lotw_cert.pem in /CardSat/ — make them from your TQSL .p12 with tools/lotw_cert_converter.html in a browser; in Settings pick DXCC → primary (state/province/...) → secondary (county/city) + zones + IOTA, all gated pickers w/ type-to-filter) — see manual §8. **u** upload · **a** re-send all · ` back

UPLOAD TO CLOUDLOG (Log → Upload to Cloudlog)
Uploads sat QSOs to a self-hosted Cloudlog/Wavelog over WiFi (an alternative to LoTW — Cloudlog can forward to LoTW). Set **URL** + read-write **API key** + **station ID** in Settings. **u** upload · **a** re-send all · ` back

VOICE MEMOS (Log → Voice Memos)
Browse newest-first (date/time/sat/len). **ENTER** play · **n** new · **d** del · **r** refresh · record via **v** on Track. SD req'd

NOTES (Log → Notes)
Free-form text notes (.txt in /CardSat/notes/, newest-first w/ date/time). Browser: **ENTER** open · **n** new · **d**+ENTER del. Editor (commands use **Fn** so **;.,/** type): type freely, ENTER=newline, DEL=backspace · **Fn+,/Fn+/** cursor L/R · **Fn+;/Fn+.** up/down · **Fn+s** save · ` exit (auto-saves)

LORA MESSAGES (Home → Messages)
CardSat-to-CardSat broadcast chat (Cap LoRa). Same freq/SF/BW = same group. Set **region** (US 33cm / EU 70cm / JP 430) in Settings for a legal default freq. **n** write/send · **;/.** scroll+select (newest on bottom) · **o** station roster · **r** retry · **m** LoRa RX/hex monitor · ` back. **Actionable msgs** (plain text, interop): a msg with **@lat,lon** / **#SAT** / **!SAT date time** → **ENTER** opens bearing compass (dist/brg/grid) / sat detail / pre-filled sked; **#SAT** carries NORAD (**#name/norad**) so it resolves across differing names. Send for the current sat & your location: **p** position (also a presence ping) · **s** satellite · **k** sked (date→time). **Roster** (**o**): who is heard, with grid + dist/brg + signal; ENTER=compass, **p**=ping. Opt-in **Auto position reply** setting answers others' positions (loop-guarded). **Rx always on**: badge + banner (opt-in beep, **Msg notify** in Settings). **Share GP elements**: a peer's **Sats** → **L** pops an **Import satellite?** prompt (sender/name/NORAD) — **y** add/update, **n** decline; checksum-verified, experimental. Needs RadioLib build.

LORA RX / HEX MONITOR (Messages → m)
Receive/inspect **any** LoRa signal (not just sats). Config: set freq (type in **MHz**), SF, BW, CR, sync, preamble, CRC — saved across reboots. **ENTER** starts RX. Monitor: live hexdump+ASCII, RSSI/SNR, **p** pause (read a frame on a busy channel) · **;/.** scroll · **s/b/c/f** tune SF/BW/CR/step · **,/** freq · ` esc. Rx-only. Untested HW

UPDATE
k/ENTER GP+clock+AMSAT+space-wx+weather · **f** fast (GP+AMSAT+favs' TX) · **a** cache all TX (auto-reboots) · **w** WiFi only

SETTINGS
,/ change · **ENTER** edit/toggle · **s** scan WiFi · opt. 2nd WiFi (field fallback) · reset = ERASE

GP SOURCE
AMSAT / **CelesTrak** JSON-PP category / **Custom URL** · **;/.** move · **ENTER** select

ROTATOR (manual)
,/ az · **;/.** el · **s** step · **x** stop · ` back

NETWORK SERVERS
rigctld PC drives rig (VFOA=DL/B=UL) · **rotctld** PC drives the configured rotator · **rigctl** drives remote rig · **Web** opt-in mobile page (no auth, trusted LAN): live sky plot, Doppler readout, tap-to-copy freqs, visible-pass list + AOS alerts, radio/rotator control, **Files** = download-only /CardSat browser (no upload)

EDIT
type · **DEL** backspace · **ENTER** ok · ` cancel

ABOUT
Build/version, IP, free heap and diagnostics (read-only). **r** Station readiness checklist · **t Tools** (55): scientific/graphing/programmer calculators, **Tiny BASIC** (0.9.59: SATSEL/LPRINT/gfx; no INPUT), **location converter** (grid/DMS/DDM/Plus/UTM/MGRS), DXCC/CQ/ITU lookups, RF/antenna workbench, link budget, phasing/stub, attenuator, RF exposure, orbit lifetime, **State vector** → **GP**, Q-codes/phonetics/RST, CTCSS, **Telnet** (0.9.67: LAN terminal, 10 saved hosts, opt. printing); **0.9.59 sat/build tools**: conjunction screener, orbital neighborhood, transponder planner, link-margin curve, debris-group screen, Doppler budget, cascade NF/G/T, sun-noise G/T, helix, L/Pi/T match, pointing loss, IMD, microstrip Z0, toroid, delta-v, thermal, Faraday, ampacity, PLL plan · **p Print**: 29 reports to network printer / serial / 80-col /Reports file (any mix); contextual **p** on report screens + all form tools, **P** all-passes & polar map · **l** License & credits · **z Games menu**: six mini-games (**;/.** pick, ENTER launch).